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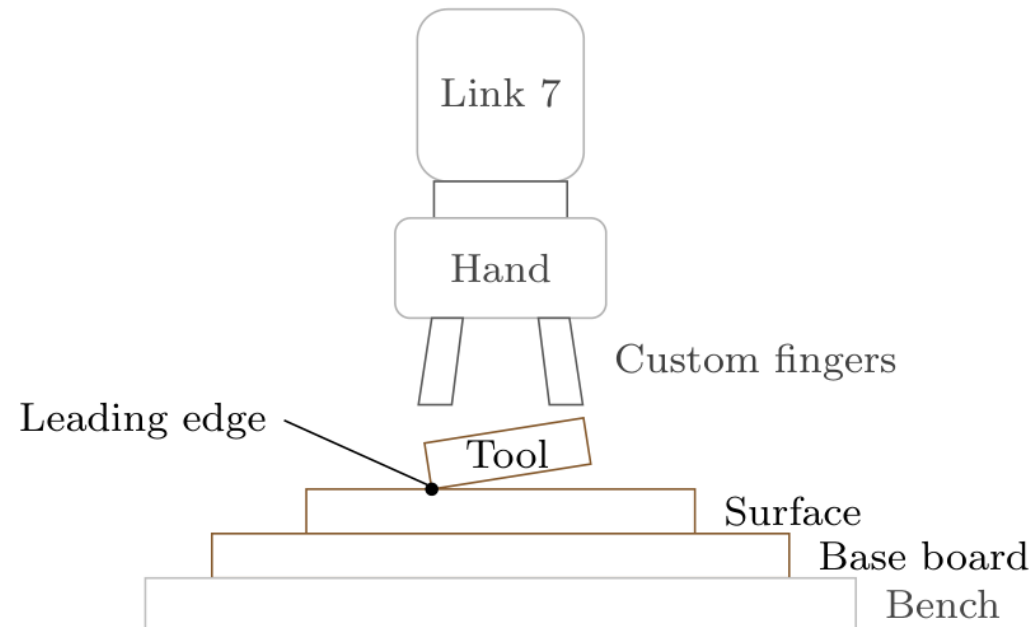
Design and Implementation of a Real-Time Impedance Controller for a 7-DOF Collaborative Robot

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Problem

- Tilted tool → edge-first contact
- Rigid tracking resists corrective rotation

Main thesis question

- **How do stiffness and the compliance centre affect passive alignment?**

Idea

- Apply the press; let contact rotate the tool

1. Cartesian impedance
2. Cartesian impedance torque
3. Surface-relative compliance
4. Centre of compliance (CoC)
5. Null-space control
6. Experimental procedure
7. Angular quantities
8. Baseline contact response
9. Effect of stiffness
10. Effect of CoC position
11. Contact response and interaction moment
12. Secondary study: null-space motion
13. Conclusion

Translational spring–damper

$$f = K_p e_p + D_p (\dot{p}_d - \dot{p}_{EE})$$

• f, m commanded force / moment

• e_p, e_R position / orientation errors

• K_p, K_R translational / rotational stiffness

• D_p, D_R translational / rotational damping

• $\dot{p}, \omega$ linear / angular velocity

• d, EE desired / measured end effector

Rotational spring–damper

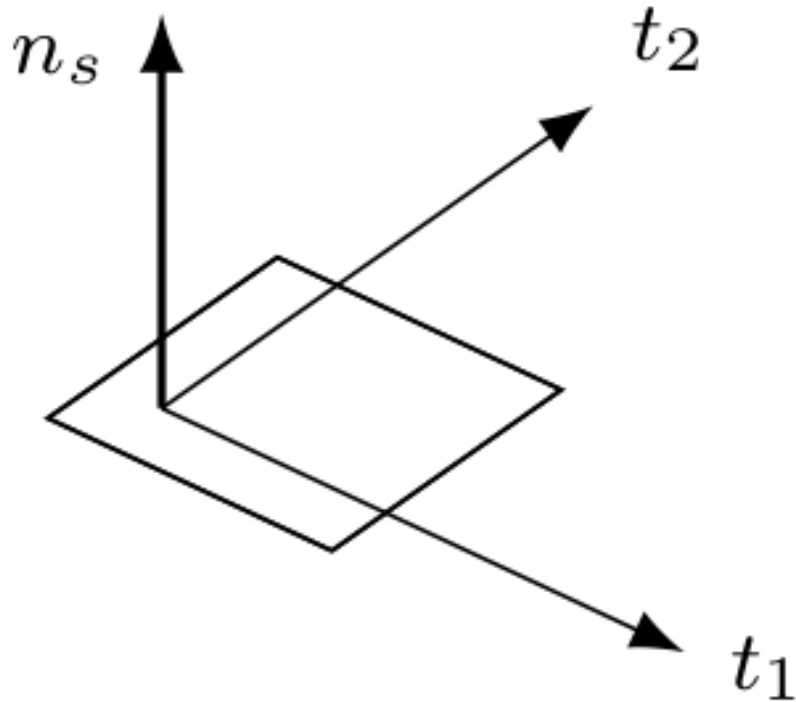
$$m = K_R e_R + D_R (\omega_d - \omega_{EE})$$

- Stiffness resists pose errors
- Damping reduces oscillation
- Compliant contact instead of rigid tracking

The Cartesian wrench is mapped to seven joint torques.

$$F = \begin{bmatrix} f \\ m \end{bmatrix}, \quad \tau_{\text{cart}} = J^{\top}(q)F$$

F: force and moment J: geometric Jacobian (6 rows, 7 columns)



- n_s outward surface normal
- t_1, t_2 two perpendicular surface tangents

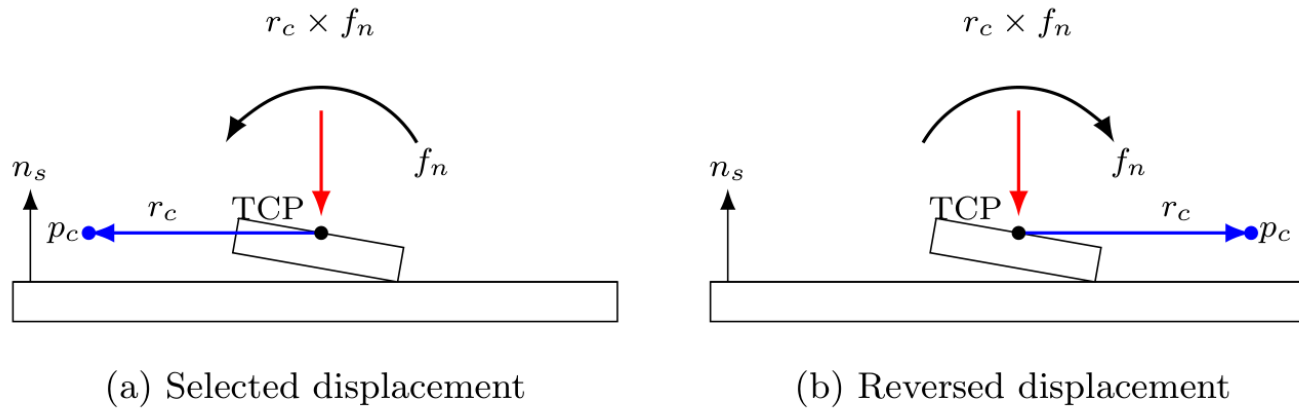
Translation

- Softer normal pressing
- Stiffer motion along the surface

Rotation

- Softer rotations about the tangents
- Stiffer rotation about the normal

Centre of compliance (CoC)



$$m = m_R + r_c \times f$$

- Tangential shift → preferred rotation
- Reversed shift → reversed additional moment

- **TCP** tool centre point; controlled reference
- p_c virtual compliance centre
- r_c displacement from TCP to centre
- f_n normal part of commanded force
- m_R rotational spring–damper moment

7 joint coordinates control a 6-DOF tool pose: 3 position + 3 orientation.

$$\dim(\ker J) = 7 - \text{rank}(J) = 7 - 6 = 1$$

At a nonsingular configuration: one independent redundant motion direction.

This freedom is a coordinated motion of the joints, not one designated joint.

$$J(q) \dot{q}_{\text{null}} = 0, \quad J \in \mathbb{R}^{6 \times 7}$$

J maps joint velocity to tool velocity: null-space motion gives zero tool velocity.

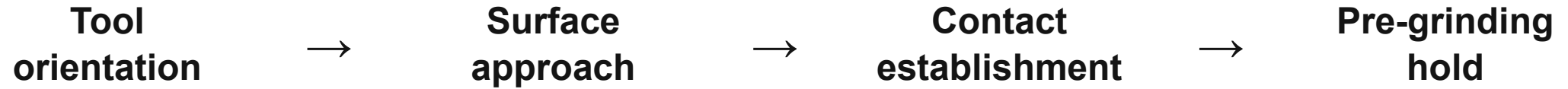
Two torque components control this internal motion:

$$\tau_{\text{null}} = \tau_d + \tau_{\sigma}$$

Damping torque: counters null-space joint motion.

Conditioning torque: steers towards a configuration with better motion capability.

Experimental procedure

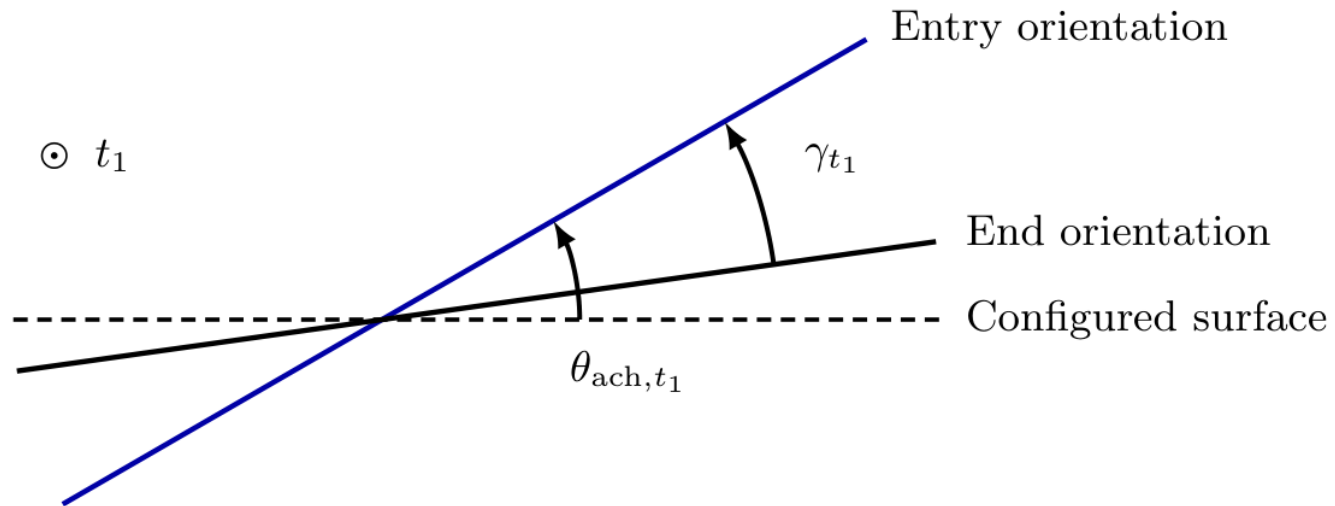


Contact study

- Rectangular tool: 40 × 120 mm
- Five-second contact phase
- Fixed entry orientation reference

Controlled comparison

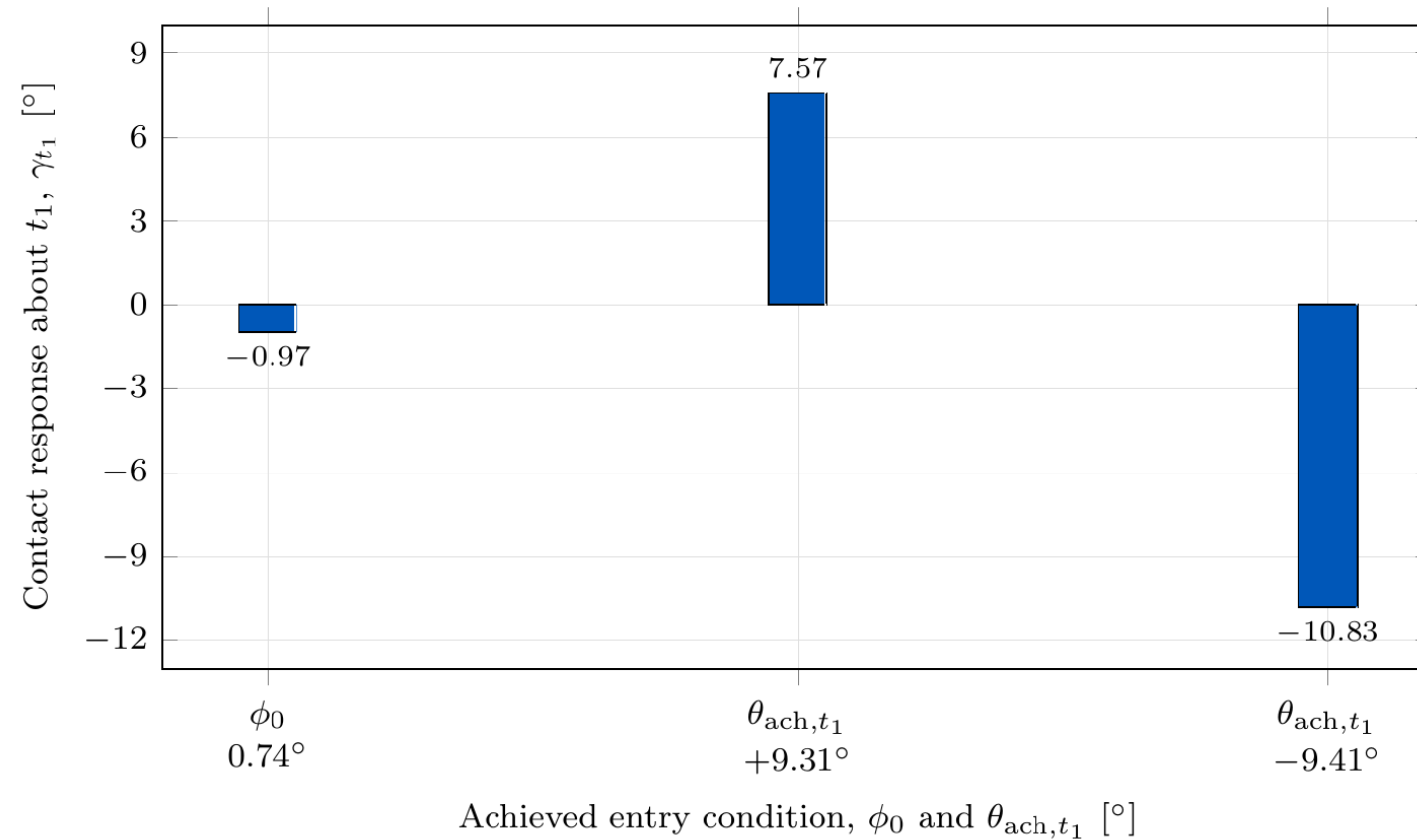
- A: centre at the TCP
- B / C: rotational / translational stiffness
- D: tangential compliance-centre position
- 57 trials; three repetitions per setting



- θ_{ach,t_1} achieved tilt at contact entry
- γ_{t_1} signed contact response about the first tangent

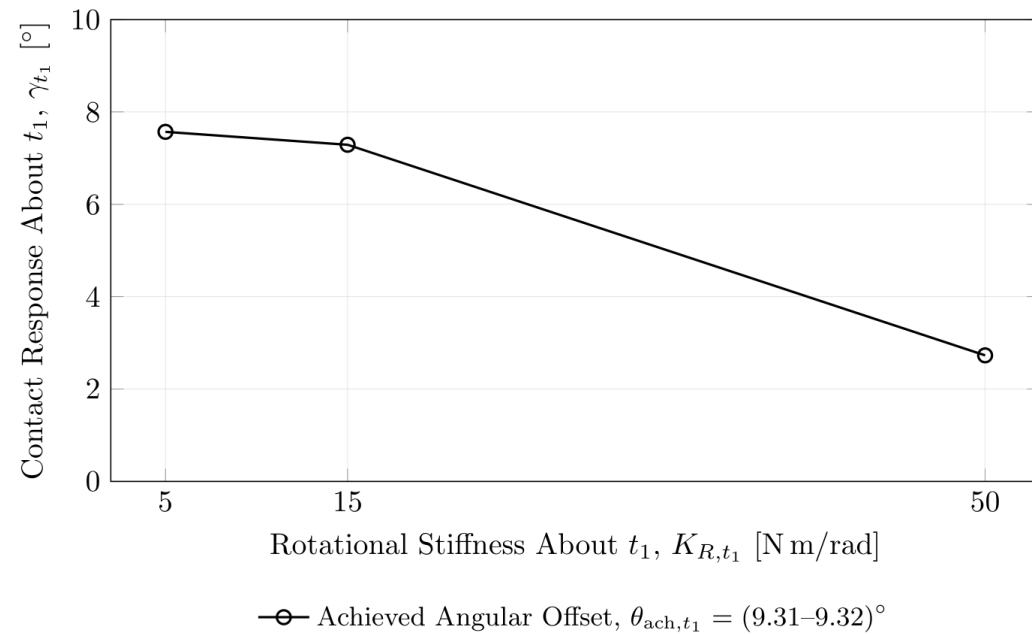
- Alignment: response has the same sign as the entry tilt
- Angles calculated from measured end-effector orientation

Baseline contact response

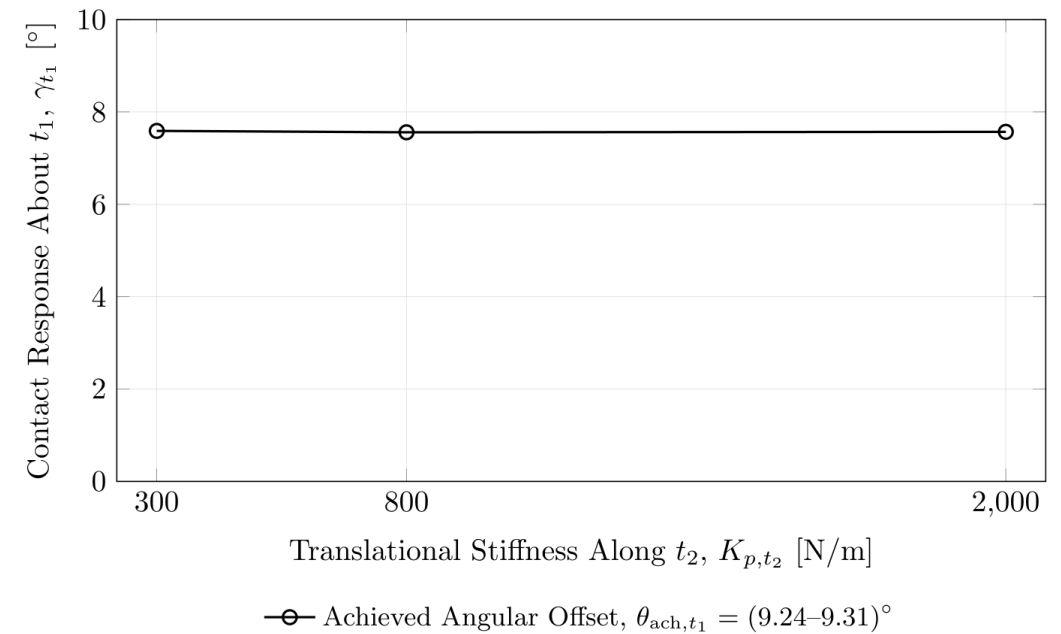


- Centre of compliance at the TCP
- Alignment response for both tilt signs
- Unequal response magnitudes

Effect of stiffness

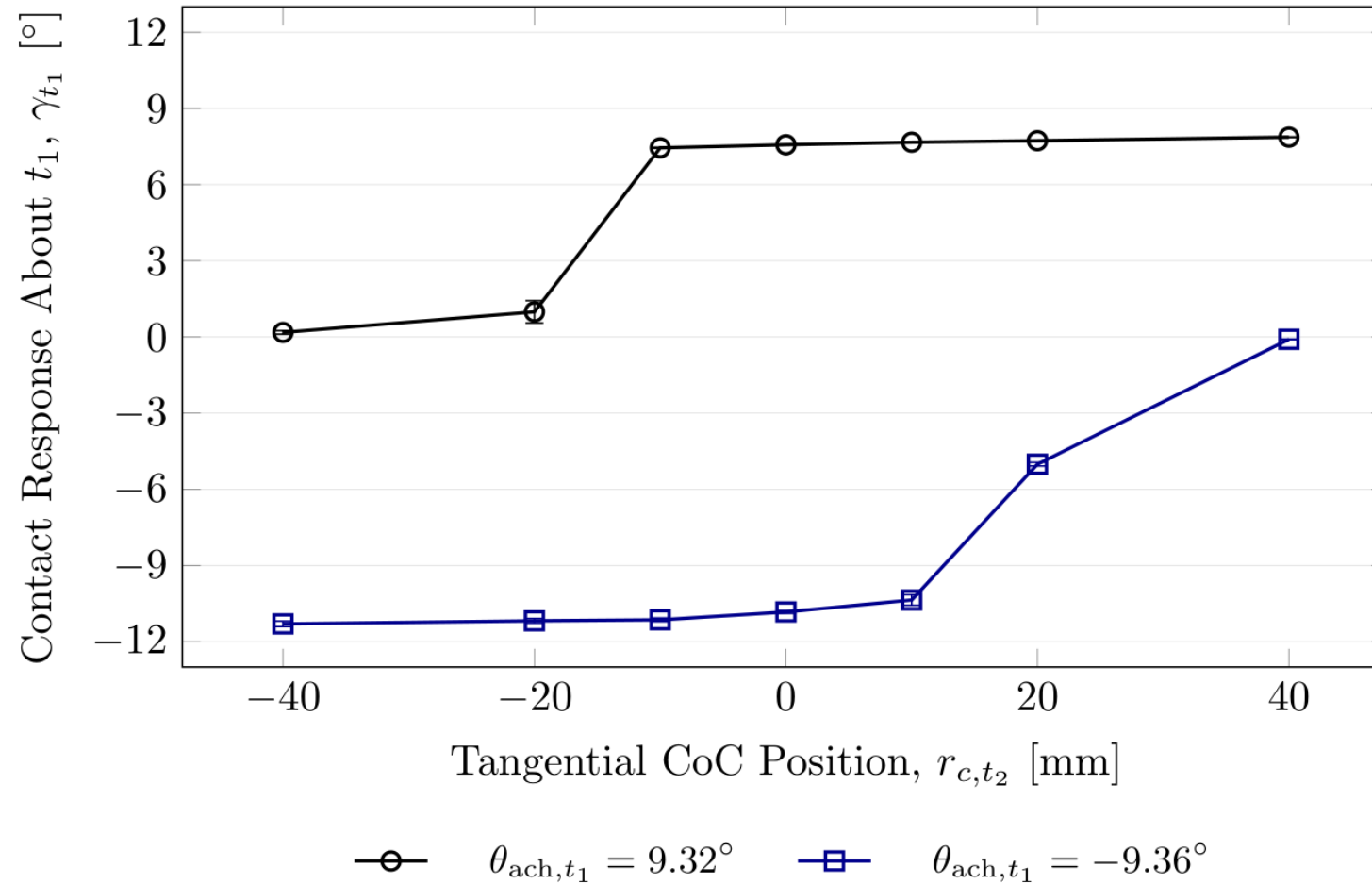


- **Rotational stiffness: 63.9% lower response**



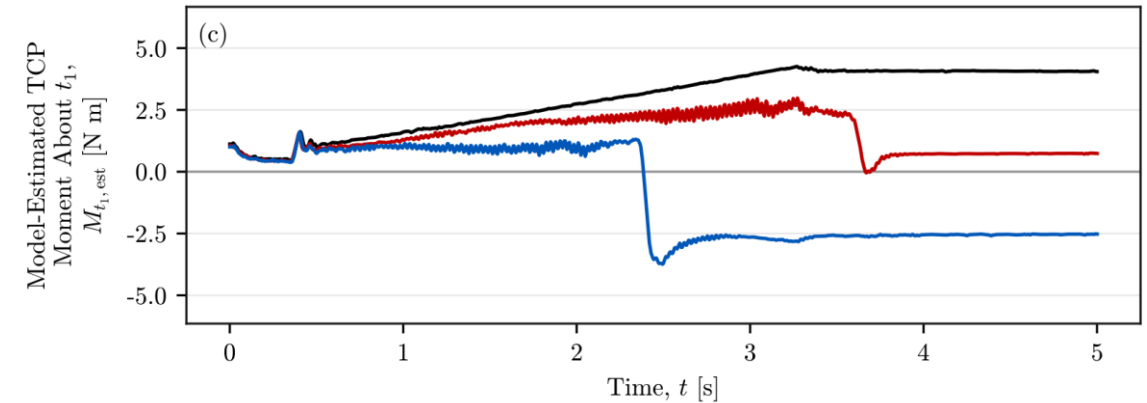
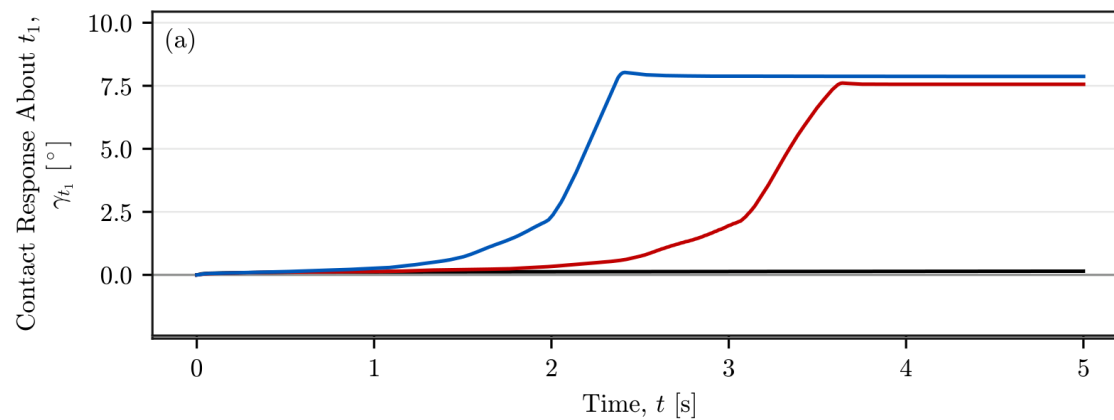
- **Tangential stiffness: 0.4% response span**

Effect of CoC position



- Selected side: about 4% higher response
- Opposite side: 98–99% lower response
- Required side reverses with the entry tilt

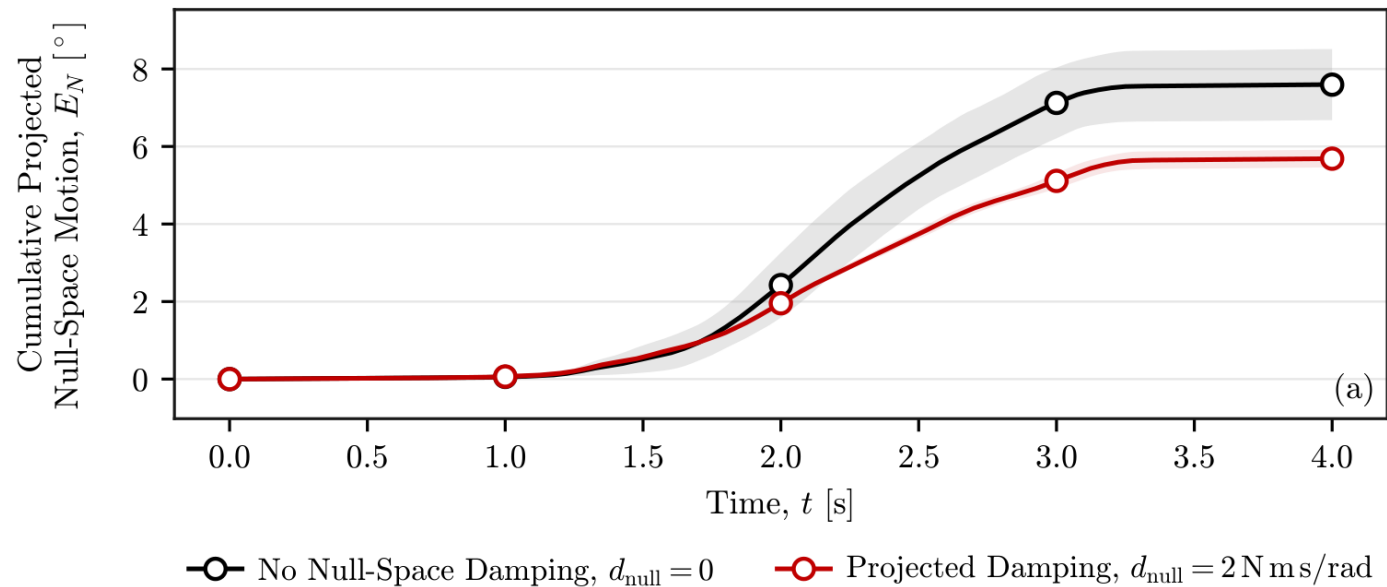
Contact response and interaction moment



— CoC Position, $r_{c, t_2} = -40$ mm — CoC at TCP, $r_{c, t_2} = 0$ — CoC Position, $r_{c, t_2} = 40$ mm

- Similar normal-force levels across the three positions
- Moment reversal; selected side reaches its steady response about 1.3 s earlier

Secondary study: null-space motion



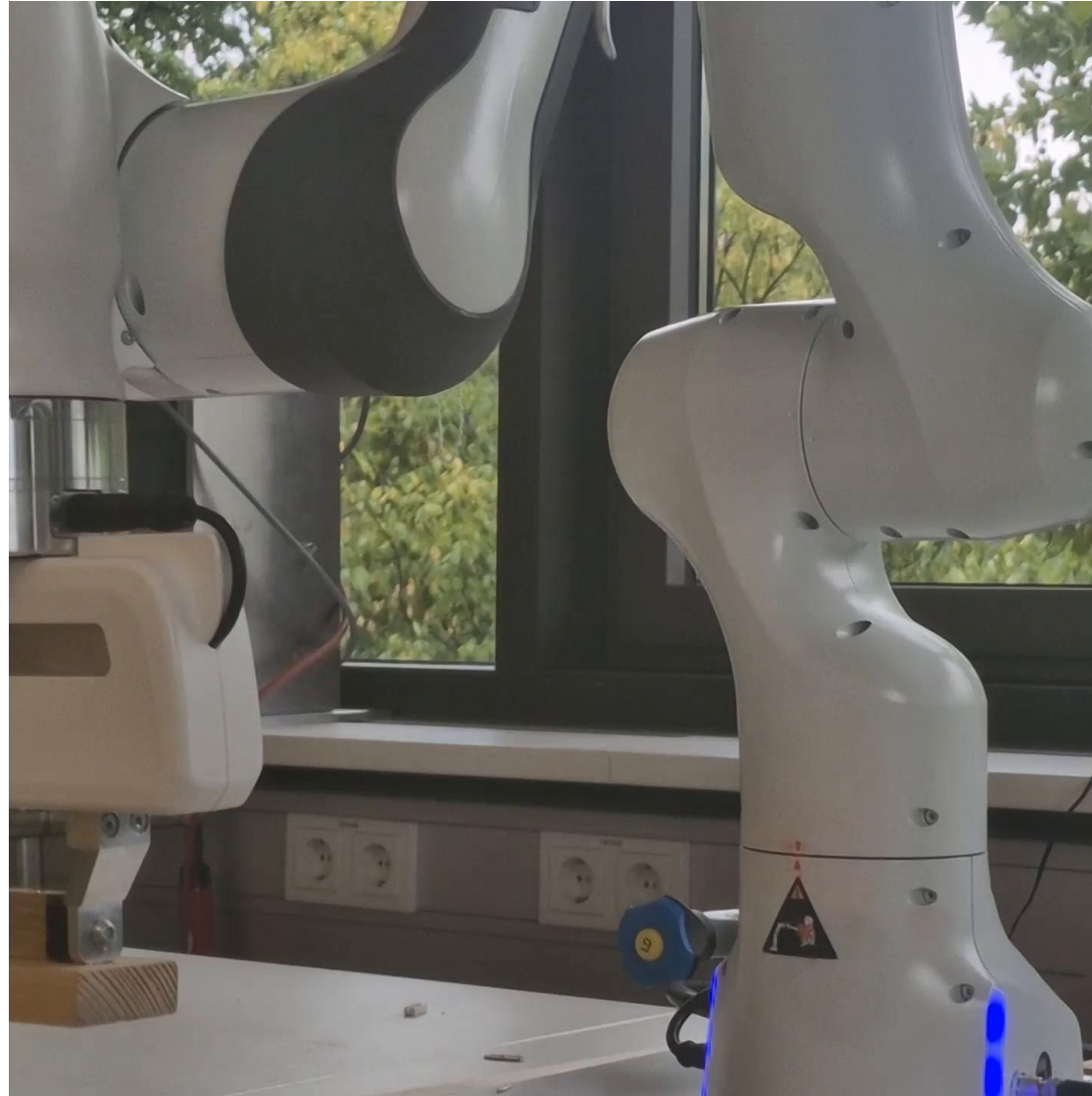
Null space

- Extra joint motion while the tool pose is held
- Damping: about 25% less total motion
- Conditioning: almost zero final drift
- Stronger conditioning: more back-and-forth motion

Contact demonstration



Disturbance demonstration



Conclusion

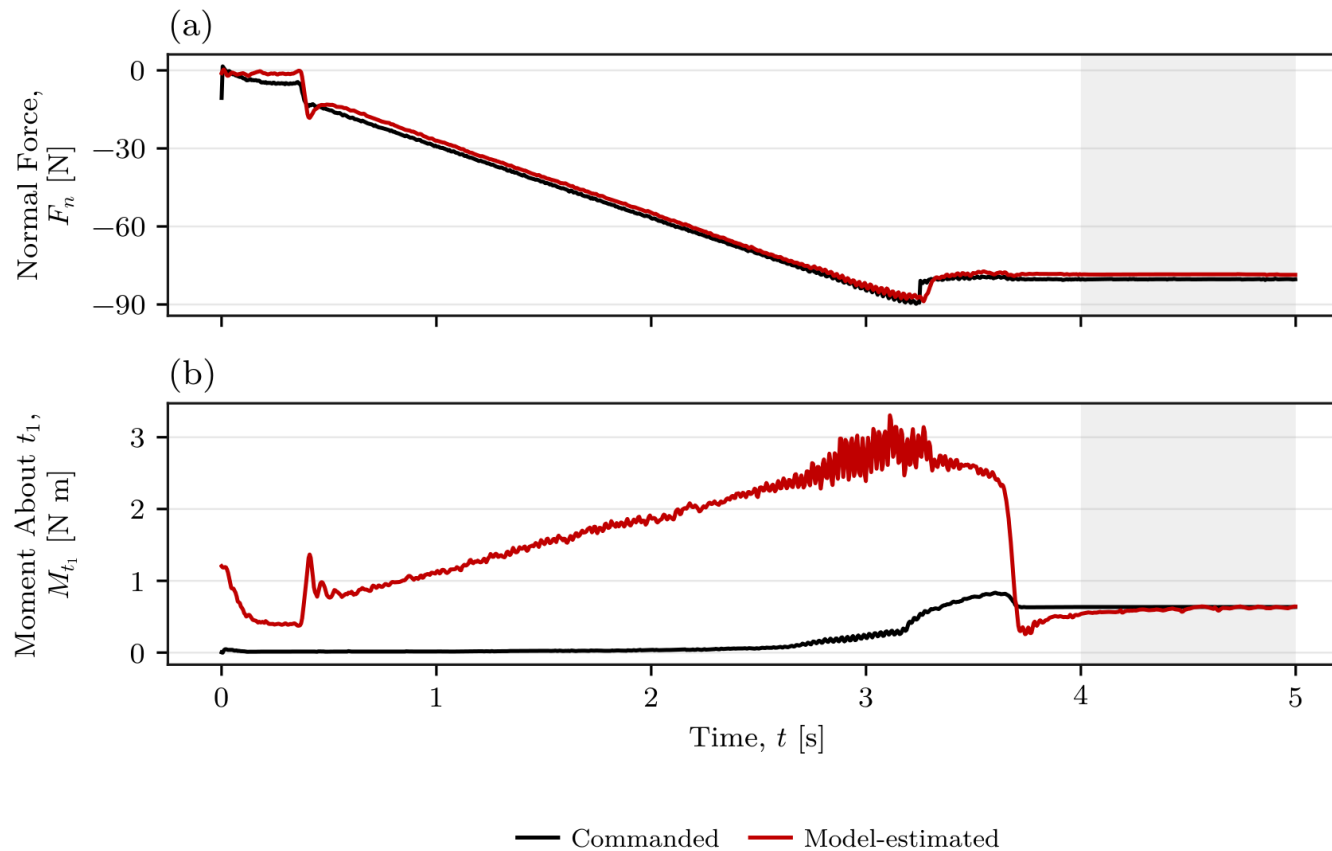
- Cartesian impedance enabled passive alignment of the rectangular tool during contact.
- Rotational stiffness mainly set the response magnitude.
- A tangential CoC displacement mainly changed the interaction moment and selected the preferred rotation direction.

Future work

- Select the CoC direction from the measured entry tilt.
- Return the CoC to the TCP after alignment.
- Extend the study to sustained grinding and independent tool-angle measurement.

Thank you

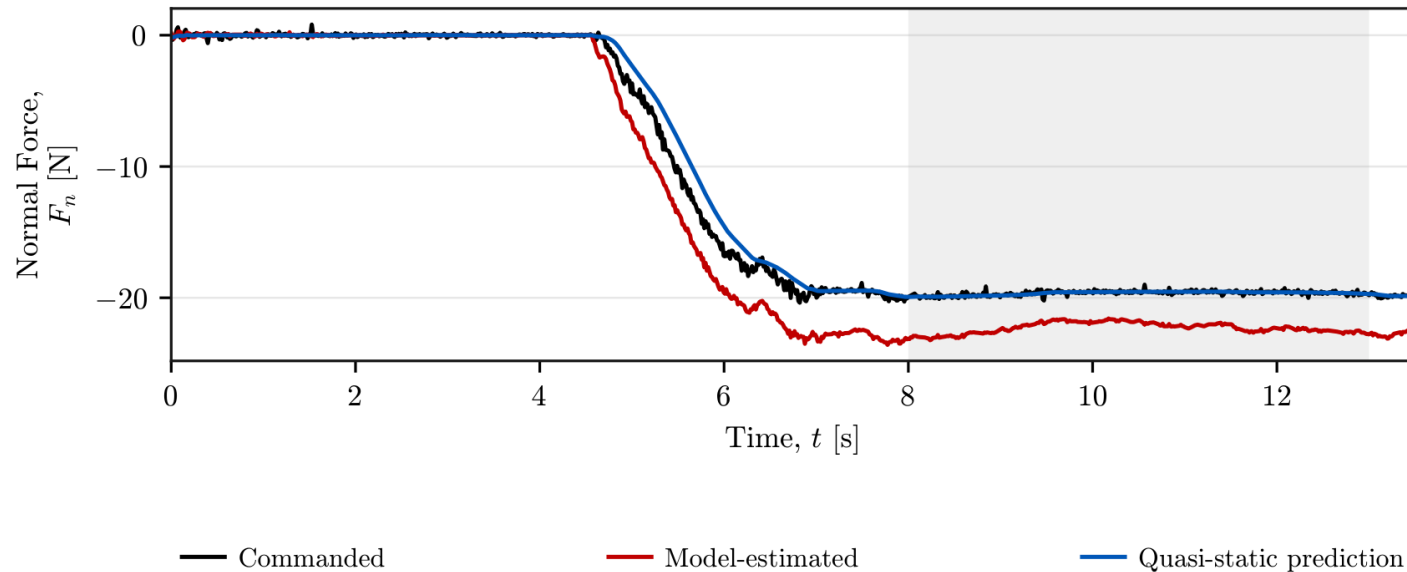
Commanded and estimated wrench



- F_n commanded normal-force component
- M_{t_1} commanded moment about the first tangent
- est model estimate from the robot
- Stationary force difference: 2.32%
- Stationary moment difference: 4.87%
- Supplementary plausibility evidence

Normal-force plausibility assessment

$$F_{n,qs} = K_{p,n} n_s^\top (e_p - e_{p,0})$$

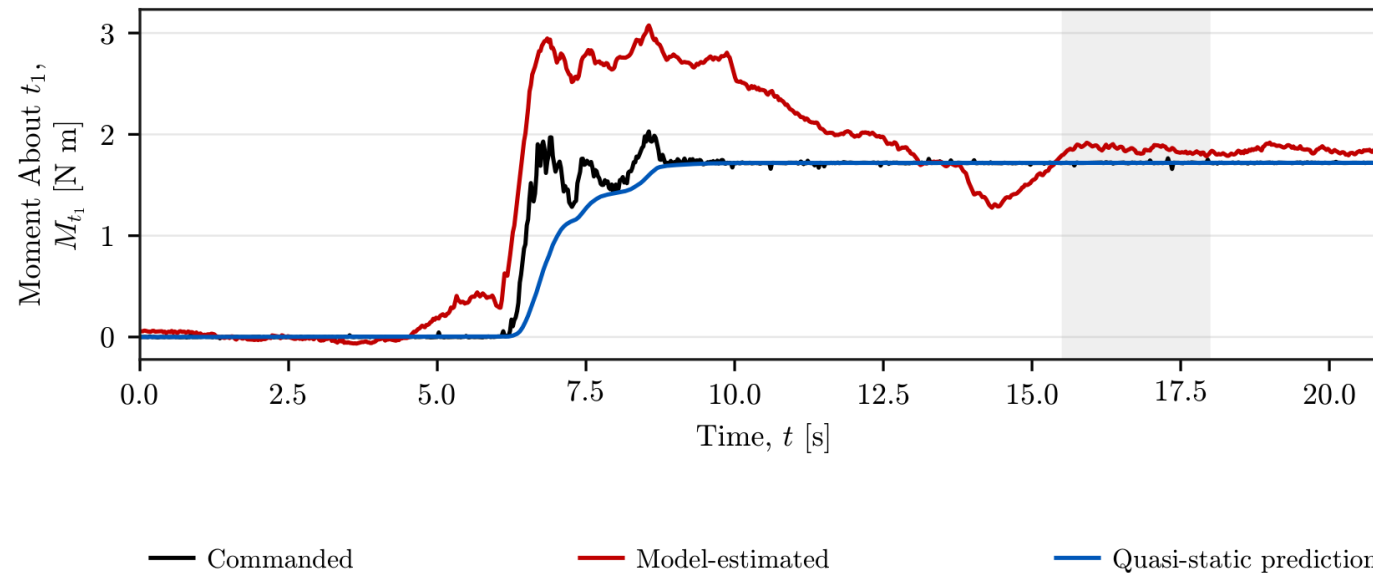


- $K_{p,n}$ normal translational stiffness
- $e_{p,0}$ position error before perturbation
- QS quasi-static spring prediction

- Spring vs command: 0.05% difference
- Estimate vs command: 13.50% difference

$$M_{t_1,qs} = K_{R,t_1} t_1^\top (e_R - e_{R,0})$$

- K_{R,t_1} rotational stiffness about the first tangent
- $e_{R,0}$ orientation error before perturbation
- QS quasi-static spring prediction



- Spring and commanded mean: 1.719 N m
- Model-estimated mean: 1.852 N m